一、设$F_1(x,y,z)=x^3+y^3+z^3-3xyz$, $F_2(x,y,z)=x+y+z-a$, 则

$$\begin{pmatrix}\frac{dy}{dx}\\ \frac{dz}{dx}\end{pmatrix}=-\begin{pmatrix}\frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial z}\\ \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial z}\end{pmatrix}^{-1}\begin{pmatrix}\frac{\partial F_1}{\partial x}\\ \frac{\partial F_2}{\partial x}\end{pmatrix}=-\begin{pmatrix}3y^2-3xz & 3z^2-3xy\\ 1 & 1\end{pmatrix}^{-1}\begin{pmatrix}3x^2-3yz\\ 1\end{pmatrix}$$

$$=-\frac{1}{3(y+z)(y-z-x)}\begin{pmatrix}1 & 3xy-3z^2\\ -1 & 3y^2-3xz\end{pmatrix}\begin{pmatrix}3x^2-3yz\\ 1\end{pmatrix}=\begin{pmatrix}\frac{(z-x)(z+x+y)}{(y+z)(y-z-x)}\\ \frac{(x-y)(x+y+z)}{(y+z)(y-z-x)}\end{pmatrix}$$

$$\Rightarrow \frac{dy}{dx}=\frac{(z-x)(z+x+y)}{(y+z)(y-z-x)},\ \frac{dz}{dx}=\frac{(x-y)(x+y+z)}{(y+z)(y-z-x)}$$

二、① 设$C=\{(x,y)|x\geqslant 0,y\geqslant 0,x+y\leqslant 1\}$

$$\iiint_D\frac{dxdydz}{(2+x+y)^2}=\iint_C dxdy\int_0^{1-x-y}\frac{dz}{(2+x+y)^2}=\iint_C\frac{1-x-y}{(2+x+y)^2}dxdy=\int_0^1 dx\int_0^{1-x}\frac{1-x-y}{(2+x+y)^2}dy$$

$$=\int_0^1\left(\int_0^{1-x}\left(\frac{3}{(2+x+y)^2}-\frac{1}{2+x+y}\right)dy\right)dx=\int_0^1\left(3\int_{x+2}^3\frac{dy}{y^2}-\int_{x+2}^3\frac{dy}{y}\right)dx=\int_0^1\left(\frac{3}{x+2}-\ln\frac{3}{x+2}-1\right)dx$$

$$=\int_2^3\left(\frac{3}{x}-\ln\frac{3}{x}-1\right)dx=3\ln\frac{3}{2}-1-\left(x\ln\frac{3}{x}\Big|_2^3-\int_2^3 xd\left(\ln\frac{3}{x}\right)\right)=5\ln\frac{3}{2}-2$$

② 设$C_z=\{(x,y)|x\geqslant 0,y\geqslant 0,x+y\leqslant 1-z\}$

$$\iiint_D\frac{dxdydz}{(2+x+y)^2}=\int_0^1 dz\iint_{C_z}\frac{dxdy}{(2+x+y)^2}=\int_0^1 dz\int_0^{1-z}dx\int_0^{1-z-x}\frac{dy}{(2+x+y)^2}=\int_0^1 dz\int_0^{1-z}\left(\frac{1}{z-3}+\frac{1}{x+2}\right)dx$$

$$=\int_0^1\left(\frac{1-z}{z-3}+\ln\frac{3-z}{2}\right)dz=5\ln\frac{3}{2}-2$$

三、(1)、$\int_1^2 dx\int_{2-x}^{\sqrt{2-x}}f(x,y)dy=\int_0^1 dy\int_{2-y}^2 f(x,y)dx+\int_1^{\sqrt{3}}dy\int_{\frac{y^2+1}{2}}^2 f(x,y)dy$

(2)、$\int_0^1 dx\int_0^{1-x}dy\int_0^{x+y}f(x,y,z)dz=\int_0^1 dy\int_0^{1-y}dx\int_0^{x+y}f(x,y,z)dz=\int_0^1 dy\int_0^y dz\int_0^{1-y}f(x,y,z)dx$

$$+\int_0^1 dy\int_y^1 dz\int_{z-y}^{1-y}f(x,y,z)dx=\int_0^1 dz\int_z^1 dy\int_0^{1-y}f(x,y,z)dx+\int_0^1 dz\int_0^z dy\int_{z-y}^{1-y}f(x,y,z)dx$$

四、设$P=x^2\ln y$, $Q=\frac{x^3}{3y}$, $\Gamma':y=-x+5(1\leqslant x\leqslant 4)$

由$\frac{\partial P}{\partial y}=\frac{\partial Q}{\partial x}=\frac{x^2}{y}$知$\int_\Gamma x^2\ln y\,dx+\frac{x^3}{3y}dy=\int_{\Gamma'}x^2\ln y\,dx+\frac{x^3}{3y}dy=\int_1^4\left(x^2\ln(5-x)-\frac{x^3}{3(5-x)}\right)dx$

$$=\frac{1}{3}x^3\ln(5-x)\Big|_1^4=-\frac{2}{3}\ln 2$$

五、设 $F(x,y,z)=f(x,y,z)+\lambda_1(x+y+z-12)+\lambda_2(x^2+y^2+z^2-56)$

$$=(\lambda_1+1)x+(\lambda_1+3)y+(\lambda_1+5)z+\lambda_2x^2+\lambda_2y^2+\lambda_2z^2-12\lambda_1-56\lambda_2$$

令 $\frac{\partial F}{\partial x}=\lambda_1+1+2\lambda_2x=0$, ①

$\frac{\partial F}{\partial y}=\lambda_1+3+2\lambda_2y=0$, ②

$\frac{\partial F}{\partial z}=\lambda_1+5+2\lambda_2z=0$ ③

①+②+③得 $\lambda_1+8\lambda_2+3=0$ ④

④代入①、②、③得 $x=\frac{4\lambda_2+1}{\lambda_2}$, ⑤

$y=4$, ⑥

$z=\frac{4\lambda_2-1}{\lambda_2}$ ⑦

⑤⑥⑦代入 $x^2+y^2+z^2=56$ 得 $\begin{cases}\lambda_2=\frac{1}{2}\\ x=6\\ y=4\\ z=2\end{cases}$ 或 $\begin{cases}\lambda_2=-\frac{1}{2}\\ x=2\\ y=4\\ z=6\end{cases}$

由连续函数在有界闭集上必然取到最小值和最大值，$f(6,4,2)=28$ 和 $f(2,4,6)=44$ 知 $f(x,y,z)$ 在前述条件下的最大值为44

六、证明：由Gauss公式，$\iiint_\Omega(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z})(x,y,z)dxdydz=0$

设 $g(x,y,z)=(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}-f)(x,y,z)$,

由积分平均值定理得存在 $(x_0,y_0,z_0)\in\overline{\Omega}$ 使得

$$\iiint_\Omega g(x,y,z)dxdydz=g(x_0,y_0,z_0)V(\Omega),\ 即$$

$$(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z})(x_0,y_0,z_0)=f(x_0,y_0,z_0)-\frac{1}{V(\Omega)}\iiint_\Omega f(x,y,z)dxdydz \qquad \square$$

七、证明：

(1)、$f(x+h)-f(x)=Jf(x)h+r(h)$，其中 $\lim_{h\to 0}\frac{\|r(h)\|}{\|h\|}=0$

$<f(x+h)-f(x),h>\geq\alpha\|h\|^2$

$\Rightarrow \langle Jf(x)h,h\rangle+\langle r(h),h\rangle \geq \alpha\|h\|^2$

利用Cauchy-Schwarz不等式得 $\langle Jf(x)h,h\rangle \geq \alpha\|h\|^2-\langle r(h),h\rangle$

$\geq \alpha\|h\|^2-\|r(h)\|\|h\|$

对于 $\varepsilon=\frac{1}{2}\alpha$, $\exists\delta>0$, $h\in B_\delta(\vec{0})$ 时 $\frac{\|r(h)\|}{\|h\|}<\varepsilon$

$\Rightarrow \langle Jf(x)h,h\rangle \geq (\alpha-\varepsilon)\|h\|^2=\frac{\alpha}{2}\|h\|^2>0$

若 $\det Jf(x)=0$, 则 $Jf(x)$ 有特征值0, 设 $h\neq 0$ 为对应的特征向量, 则 $Jf(x)h=0$,

取 $x>0$ 充分大使得 $\frac{h}{x}\in B_\delta(\vec{0})$, 则 $\langle Jf(x)\frac{h}{x},\frac{h}{x}\rangle=0$, 矛盾

$\Longrightarrow \det Jf(x)\neq 0$

(2)、由 $f\in C^1(\mathbb{R}^n)$ 及 $\det f(x)\neq 0$ 可得 $f(\mathbb{R}^n)$ 为开集

对 $\forall x\neq 0\in\mathbb{R}^n$, $\alpha\|x\|^2\leq\langle f(x)-f(0),x\rangle\leq\|f(x)-f(0)\|\|x\|$

$\Rightarrow \alpha\|x\|\leq\|f(x)-f(0)\|\leq\|f(x)\|+\|f(0)\|$

$\Rightarrow$ 若点列 $\{f(x_i)\}$ 有界, 即存在 $r>0$ 使得 $\|f(x_i)\|<r$, 则 $\|x_i\|<\frac{r+\|f(0)\|}{\alpha}$, 则点列 $\{x_i\}$ 有界

取 $f(\mathbb{R}^n)$ 中任一收敛点列 $\{f(x_i)\}$, 由于 $\{f(x_i)\}$ 收敛, 故 $f(x_i)$ 有界, 则 $\{x_i\}$ 有界, 由Bolzano-Weierstrass定理, 选出 $\{x_i\}$ 中收敛子列 $\{x_{k_i}\}$, 设 $\lim\limits_{i\to\infty}x_{k_i}=x$, 由 f 的连续性知 $\lim\limits_{i\to\infty}f(x_{k_i})=f(x)=\lim\limits_{i\to\infty}f(x_i)$

$\Rightarrow \lim\limits_{i\to\infty}f(x_i)\in f(\mathbb{R}^n)$

$\Rightarrow f(\mathbb{R}^n)$ 是闭集

因此 $f(\mathbb{R}^n)^C$ 是开集, 而 $\mathbb{R}^n=f(\mathbb{R}^n)\cup f(\mathbb{R}^n)^C$, 但是 $\mathbb{R}^n$ 是连通集, 不能分解成两个不相交的非空开集之并, 故 $f(\mathbb{R}^n)$ 与 $f(\mathbb{R}^n)^C$ 中有一个是空集, 故 $f(\mathbb{R}^n)=\mathbb{R}^n$ □

八、证明: 设 $\varphi:\mathbb{R}^2\longrightarrow\mathbb{R}^2;(r,\theta)\longmapsto(r\cos\theta,r\sin\theta)$, 则 $\frac{\partial(f\circ\varphi)}{\partial r}=\frac{\partial f}{\partial x}\cos\theta+\frac{\partial f}{\partial y}\sin\theta$

$$\lim_{\varepsilon\to 0^+}\iint_{\varepsilon^2\leq x^2+y^2\leq 1}\frac{1}{x^2+y^2}\left(x\frac{\partial f}{\partial x}+y\frac{\partial f}{\partial y}\right)(x,y)\,dxdy=\lim_{\varepsilon\to 0^+}\int_0^{2\pi}d\theta\int_\varepsilon^1\frac{1}{r^2}\left(r\cos\theta\frac{\partial f}{\partial x}+r\sin\theta\frac{\partial f}{\partial y}\right)rdr$$

$$=\lim_{\varepsilon\to 0^+}\int_0^{2\pi}d\theta\int_\varepsilon^1\frac{\partial(f\circ\varphi)}{\partial r}dr=\lim_{\varepsilon\to 0^+}\int_0^{2\pi}\left((f\circ\varphi)(1,\theta)-(f\circ\varphi)(\varepsilon,\theta)\right)d\theta=\lim_{\varepsilon\to 0^+}\int_0^{2\pi}-(f\circ\varphi)(\varepsilon,\theta)\,d\theta$$

由积分平均值定理知存在 $\theta_0\in[0,2\pi]$ 使得 $\int_0^{2\pi}-(f\circ\varphi)(\varepsilon,\theta)\,d\theta=-2\pi(f\circ\varphi)(\varepsilon,\theta_0)$

由 f 的连续性知 $\lim\limits_{\varepsilon\to 0^+}\int_0^{2\pi}-(f\circ\varphi)(\varepsilon,\theta)\,d\theta=-2\pi\lim\limits_{\varepsilon\to 0^+}(f\circ\varphi)(\varepsilon,\theta_0)=-2\pi f(0,0)$ □